Period lattices and marked affine quotients

human_compression project — private author draft

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1. Purpose and retained interfaces

A finite list of affine words can describe infinitely many finite marked groups. The useful information is the translation lattice, together with the few translation differences arising when two distinguished subgroups have the same linear part. We make this observation into an exact presentation and intersection criterion. The accompanying application uses it twice: for the three cubic families of type $\{6,4\}$ and for three families of type $\{6,6\}$ in Hou–Feng–Leemans [HFL]. The latter calculation is reused from the earlier proof in this project, not performed again as new research.

A marking is an ordered tuple of nonidentity involutions. For a subset I of its indices, G_I denotes the subgroup on those markers. The intersection property is $G_I \cap G_J = G_{I\cap J}$. We retain the standard regular-polytope correspondence. The following short argument is reused from [OK, Lemma 2.1], and included here to close this paper's interface.

Rank-three test. For three distinct involutions with commuting endpoints and adjacent products of order at least three, IP is equivalent to $G_{01} \cap G_{12} = G_1$.

Proof. Necessity is immediate. For sufficiency, the endpoint subgroup is $G_{02} = \{1, r_0, r_2, r_0r_2\}$. If $G_{02} \cap G_{01}$ contains r_2 or r_0r_2, then $r_2 \in G_{01} \cap G_{12} = G_1$, contradicting distinctness. Thus this intersection is G_0; dually $G_{02} \cap G_{12} = G_2$. These equalities exclude every generator from the subgroup on the other two. Intersections involving singletons now follow, and intersections involving the empty or full index set are immediate. This covers all subset pairs. □

The compatible-product construction used below is the dihedral instance of the supported-kernel theorem [UA, Theorem UA.SK]. We give its rank-three specialization explicitly so that its limited hypotheses can be checked in an application. The general kernel and deletion arguments remain attributed to their earlier sources and the reviewed umbrella. No historical priority is asserted for affine lattices or fiber products.

2. Recognizing an affine presentation

We write an integral affine transformation as (M, t), acting by $v \mapsto Mv + t$, with multiplication

$$(M, t)(N, u) = (MN, t + Mu). \tag{1}$$

Let E be a group generated by finitely many such transformations, and let $Q \le \mathrm{GL}_d(\mathbb{Z})$ be its finite linear image. Its translation kernel L is a subgroup of $\mathbb{Z}^d$, invariant under Q. A translation

is written $t_v = (I, v)$.

Lemma 1 (fixed Schreier lattice). Choose a word lift $\hat{q} \in E$ for each $q \in Q$, with $\hat{1} = 1$. Then L is generated by the translations

$$\widehat{\overline{s}q}^{-1} s\hat{q}, \qquad s \text{ a distinguished generator}, \quad q \in Q. \tag{2}$$

Proof. The products have trivial linear image. Write a partial word as $\hat{q}n$, with $n \in L$. Left multiplication by s changes this to $\widehat{\overline{s}qt}n$, where t is the transition in (2). Induction expresses a word of trivial linear image as a product of these transitions; its final representative is $\hat{1} = 1$. Every transition itself is in L, so they generate exactly L. □

The normalization in (2) matters. The raw differences $s\hat{q}\widehat{\overline{s}q}^{-1}$ need not generate L without conjugation. Each is a conjugate of the corresponding transition by $\widehat{\overline{s}q}$. Consequently their Q-orbits generate L. In particular, if a proposed Q-invariant lattice contains every raw difference, it contains every normalized transition and hence L. If exhibited raw differences also span the proposed lattice, equality follows. This invariant-lattice version justifies the raw-difference format of the unchanged earlier $\{6, 6\}$ checker [C66]. The new $\{6, 4\}$ checker uses (2) directly.

This is a finite proof of a lattice identity. Containment in a proposed lattice is checked on the normalized transitions, or on the raw differences with the stated invariance condition. Equality follows when independent integer combinations give a basis of that lattice, or when the greatest common divisor of the full-rank minors gives its exact index. Sampling finite quotients is not a replacement for this step.

Lemma 2 (presentation bridge). Let U be a finitely presented marked group whose relators hold on the affine generators of E. Suppose their linear images define the same epimorphism to Q. If an exact Reidemeister–Schreier presentation proves $\ker(U \to Q) \cong \mathbb{Z}^d$, and the affine translation kernel L has rank d, then the marked epimorphism $U \to E$ is an isomorphism.

Proof. The restriction maps $\ker(U \to Q)$ onto L: an affine translation has a preimage in U, and its linear image is trivial. This is a surjection between free abelian groups of rank d. Tensoring with $\mathbb{Q}$ shows that its kernel has rank zero. A subgroup of a free abelian group is torsion-free, so that kernel is trivial. The remaining kernel of $U \to E$ lies in this subgroup, and is therefore trivial as well. □

The hypothesis concerns the actual kernel presentation. Computing only its abelianization would not establish it: a nonabelian group can have free abelian abelianization of rank d. Our applications retain the exact rewriting and Tietze algorithms used to verify this finite presentation step. It is a computer-assisted input, not a Lean proof.

3. Periods and all-parameter intersections

Let $w_1, \dots, w_d \in U = E$ be translation words with linearly independent vectors $v_1, \dots, v_d$. Write $\Lambda = \sum_i \mathbb{Z} v_i \le L$ and suppose it is Q-invariant. For a positive integer m, adjoining $w_i^m = 1$ to the complete presentation of U gives exactly

$$E_m = E/(m\Lambda). \tag{3}$$

Indeed, the normal closure of the powers is $m\Lambda$: conjugation by an affine element acts on translations by its linear part, and this lattice is already invariant. Hence

$$|E_m| = |Q|[L : \Lambda]m^d. \tag{4}$$

This proves equality for the presented group, not merely the order of a chosen quotient. If Q is solvable, every E_m is solvable.

Suppose now that the marking has rank three with distinct generators, its endpoints commute, both adjacent products have order at least three, and its two adjacent subgroups $A = E_{01}$ and $B = E_{12}$ are finite and inject into Q. For each $q \in \overline{A} \cap \overline{B}$, let $(q, t_A(q))$ and $(q, t_B(q))$ be their unique affine lifts. Put

$$\Delta(q) = t_A(q) - t_B(q). \tag{5}$$

Theorem 3 (period-lattice intersection criterion). The images of A and B in E_m meet precisely in the elements indexed by

$$\{q \in \overline{A} \cap \overline{B} : \Delta(q) \in m\Lambda\}. \tag{6}$$

In particular, if $A \cap B = \langle r_1 \rangle$ and every other value $\Delta(q)$ lies outside Λ, then every E_m, $m \geq 1$, has the full rank-three intersection property and the same adjacent orders as E. Its order is (4).

Proof. Equality of two cosets modulo translations first forces equality of their linear parts. For equal linear part, their affine quotient is translation by the difference (5). This proves (6). Each adjacent subgroup remains injective in the quotient because its linear action is faithful, so its order and the product order of its two reflections are unchanged. The assumed exclusions apply to $m\Lambda \subseteq \Lambda$. Thus the adjacent intersection is exactly the common involution subgroup, and the rank-three test proves all subset intersections. □

The proof separates two possibilities often hidden by equal group-order counts. The point-group intersection may already have order two, as in the $\{6, 4\}$ application. Or it may be larger, with a short list of nonzero translation differences, as in the $\{6, 6\}$ application. Those differences must be excluded from the period lattice explicitly.

4. Changing one adjacent order

Let $(B; a, b, c)$ be a rank-three string C-group of type $\{h, q\}$, with h, q even, and suppose the map

$$\rho : B \longrightarrow B_{12}, \qquad a \mapsto 1, b \mapsto b, c \mapsto c \tag{7}$$

is a retraction. For k a positive multiple of q, take the standard marked epimorphism $D_{2k} \to B_{12} = D_{2q}$. The compatible product

$$F = \{(g, d) \in B \times D_{2k} : \rho(g) = \overline{d}\} \tag{8}$$

has markers $(a,1),(b,\widetilde{b}),(c,\widetilde{c})$. The evenness of h is forced by (7); it also ensures that the first pair has unchanged order h.

Theorem 4 (one-edge extension). These markers generate F. Its order is $|B|k/q$, its type is $\{h,k\}$, and it has the intersection property. The projection onto B is injective on the first adjacent subgroup. If B is solvable, so is F.

Proof. The word $(r_1r_2)^q$ generates the entire cyclic kernel of (8), since its first coordinate is trivial and its second has order k/q. Surjectivity onto B then proves generation and the order formula. The first adjacent subgroup has second coordinate of order at most two, with that coordinate determined by the image under (7); thus its projection is injective. The second adjacent subgroup is the full D_{2k}. If an element belongs to both, its projection belongs to $\langle b\rangle$ by IP in B. Subtract the appropriate common involution. The remaining element lies in the first subgroup and in the projection kernel, hence is trivial. This proves IP by the rank-three criterion. Solvability follows from the cyclic kernel. □

Theorem 4 is a specialization of UA.SK; it is included to make clear that only the indicated deletion is needed here. More simultaneous edges use the supported coordinate kernels of that theorem.

There is a useful exact presentation variant. Suppose a complete marked presentation for B contains $(bc)^q=1$. Replace that relation by $(bc)^k=1$, retain the remaining relators, and add $[a,(bc)^q]=1$. Call the resulting presented group H. If every retained relator holds in (8), then $H=F$ as a marked group. Indeed, $C=\langle(bc)^q\rangle$ is normal in H: the last relation centralizes it by a, and b,c invert it. Its order is at most k/q, and $H/C=B$ by the presentation. Consequently $|H|\le|B|k/q$. The marked surjection onto (8) attains this bound, proving equality and injectivity. The same argument is valid for an infinite base if the finite kernel maps injectively and the induced map on the base is an isomorphism.

5. Attribution and proof cost

The affine lattice method and Reidemeister–Schreier theory are classical. The source already used rank-three abelian normal subgroups and their power subgroups. The contribution of this organization is an explicit point quotient, one kernel calculation per affine group, one comparison of period lattices, and a fixed parabolic-overlap test shared by the three families. The compatible-product theorem comes from the existing supported-kernel proof library. These are proof replacements and reusable interfaces; their formulation does not establish historical priority.

The mathematical dependency includes the complete relevant written umbrella, the finite presentation programs, the exact affine arithmetic, and their declared external algorithms. A finite table is not free merely because its entries are small. The new Lean certificate in the application verifies a separate sign obstruction for one literal source presentation. It does not formalize the all-parameter affine theory in this paper. No numerical compression rate is assigned before matching the complete source and replacement obligations under one measure.

References

[HFL] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Regular 3-polytopes of order* 2^np, arXiv:2001.02945v2, 14 July 2024, 17 pp. The bound primary is `sources/regular_2np2025.pdf`.

[UA] *Supported extensions and finite two-group quotients*, the frozen Topic A umbrella, `phase207_topic_A_umbrella.md`, especially UA.K, UA.SK, UA.Affine, UA.Square and UA.Edge. Its full dependency scope is retained.

[OK] *Odd kernels, two characters, and cyclotomic polyhedra*, `phase236_odd_kernel_manuscript.md`, Lemma 2.1. The principal results of that application are attributed to Hou–Feng–Leemans–Qu.

[C66] *One affine model for all three {6,6} lattice families*, `phase2_cubic_lattices.md`, with its exact affine and presentation programs.